

For any series, in order to conclude that it converges or diverges, you MUST state which test you used, and show the work to verify every hypothesis of that test.

1. Determine, with justification, whether the following series converge or diverge.

(a)
$$\sum_{n=2}^{\infty} \frac{1}{(\ln(\ln n))^n}$$

(b)
$$\sum_{n=2}^{\infty} \frac{n^{-n} x^n}{\ln(n)}$$

2. Show that

$$\int_0^a e^{-x^2/2} dx \quad \text{is given by} \quad \sum_{n=0}^{\infty} \frac{(-1)^n a^{2n+1}}{n! 2^n (2n+1)}$$

for any value of a .

3. When a is positive, the series above is an alternating series.

- (a) Approximate

$$\int_0^1 e^{-\frac{1}{2}x^2} dx$$

to four decimal places accuracy by using the Alternating Series Test and its **remainder theorem**. (Be careful to not round intermediate steps.)

- (b) Using the same number of terms as you used in your sum in part (a), compute the midpoint approximation M_n for the integral

$$\int_0^1 e^{-x^2/2} dx$$

- (c) Which approximation is more accurate? This can and should be determined using just parts (a) and (b).

4. The series in Question 2 does not immediately satisfy the hypotheses of the Alternating Series Test when $a = 3.75$ because the magnitudes of the terms are not decreasing.

(a) Observe the index n_0 after which the magnitudes of the terms of $\sum_{n=0}^{\infty} \frac{(-1)^n 3.75^{2n+1}}{n! 2^n (2n+1)}$ do decrease. Hint: you can attempt to plot points or try numbers in your calculator.

(b) Explain how you (mathematically) know that the value of n_0 you found is correct.

(c) The portion of the series after your value of n_0 does satisfy the hypotheses of the Alternating Series Test. Using the remainder theorem, what is the smallest number of terms (of the original series) that you must use to get guarantee an error less than 0.0005? (You may want to use a computer to help you determine this.)

5. When the integral test applies, convergence of $S = \sum_{n=1}^{\infty} f(n)$ is understood in terms of convergence of the related improper integral $I = \int_1^{\infty} f(x) dx$. It is not necessarily true that the values S and I are close together. However, the discrepancy between partial sums S_n and the value S *can* be estimated using integrals.

(a) Verify carefully that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ satisfies the hypotheses of the integral test.

(b) Compute the partial sum $S_8 = \sum_{n=1}^8 \frac{1}{n^2}$ to five decimal places using a calculator.

(c) Compute the integrals $I_8 = \int_8^{\infty} \frac{1}{x^2} dx$ and $I_9 = \int_9^{\infty} \frac{1}{x^2} dx$.

(d) Since this series satisfies the hypotheses of the integral test, it can be shown that

$$I_9 \leq S - S_8 \leq I_8$$

Rearranging, find an interval that must contain the true value S . Give at least 5 decimal places for the endpoints. (Hint: the true value of S is approximately 1.644934066 in this case, so your interval should contain that number.)

6. Repeat the exact same process of question 5(b-d) for the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$.

7. Repeat the exact same process of question 5(b-d) for the series $\sum_{n=1}^{\infty} \frac{1}{n^6}$. Use at least 6 decimal places.

8. When using a Taylor polynomial to approximate a function, we expect there to be some discrepancy between the exact value of the function and the value found using our polynomial approximation. We cannot (usually) determine exactly what this error is, but it is possible to show that the (absolute value) of the error in using the n th Taylor polynomial to approximate $f(x)$ cannot exceed the bound

$$E(n, x) = \frac{M}{(n+1)!} |x - a|^{n+1}$$

where M is the maximum value of $|f^{(n+1)}(x)|$ and a is the base of our expansion.

- (a) State the value of $\sin(\pi/2)$.
 - (b) State the third Taylor polynomial for $f(x) = \sin x$ based at $x = 0$ and use it to estimate $f(\pi/2)$.
 - (c) State the exact error in using this approximation; i.e., state the difference between the exact value in (a) and the approximation in (b) to at least 4 decimal places.
 - (d) Find the value of M for this problem. That is, find the maximum value of $|f^{(4)}(x)|$ on the interval $[0, \frac{\pi}{2}]$ where $f(x) = \sin x$. You may do this by sketching a graph of $|f^{(4)}(x)|$ and then determining what the maximum is, or by using calculus techniques.
 - (e) Find the error bound $E(3, \pi/2)$ for approximating $f(\pi/2)$ using the 3rd Taylor polynomial to at least 4 decimal places.
 - (f) Verify that your answer from (c) is less than your answer from (e).
9. Let's do it again, but where exact evaluation is not so easy.
- (a) Use the third Taylor polynomial for $f(x) = \sin x$ based at $x = 0$ and to estimate $f(\pi/25)$.
 - (b) Find an **overapproximation** for the value of M for this problem. That is, find a number that is at least as big as the maximum value of $|f^{(4)}(x)|$ on the interval $[0, \frac{\pi}{25}]$ where $f(x) = \sin x$. Hint: graph the function $|f^{(4)}(x)|$ and notice that it is *always* below some number, no matter what interval we are focused on.
 - (c) Using your value of M , find an error bound $E(3, \pi/25)$ for approximating $f(\pi/25)$ using the 3rd Taylor polynomial to at least six decimal places.
 - (d) Determine an interval which is guaranteed to contain the true value of $\sin(\pi/25)$. Provide at least six decimal places for the endpoints. (Of course, you are not forbidden to check with a calculator that the value it produces for $\sin(\pi/25)$ is really in your interval.)

The following example shows how to rigorously find a Taylor series expansion. Recall the notation $f^{(n)}(x)$ for the n^{th} derivative of a function f .

10. Consider expanding the function $f(x) = xe^x$ around the point $x = 2$.

(a) Compute $f'(x)$, $f''(x)$, and $f'''(x)$. Hint: simplify as you go.

(b) Make an educated guess of the general form for $f^{(n)}(x)$ (not $f^{(n)}(2)$).

- Plug in $n = 3$ to make sure $f^{(3)}(x)$ is the same as $f'''(x)$ in (a). If not, go back.

(c) Based on your guess in (b), determine what $f^{(n+1)}(x)$ should be, i.e. replace each occurrence of n by $n + 1$.

(d) Compute $f^{(n+1)}(x)$ a different way: differentiate your guess from (b).

Note $f^{(n+1)}(x) = \frac{d}{dx} f^{(n)}(x)$.

- If the previous two steps match, then your guess for the general form for $f^{(n)}(x)$ was correct. If not, then go back and fix the earlier steps.

(e) Evaluate $f^{(n)}(2)$ for general n . Mentally verify that this matches $f(2)$, $f'(2)$, $f''(2)$. (Because of the first bullet point, you should not have to check when $n \geq 3$.)

(f) Finish finding the Taylor series for $f(x)$ centered at 2.